

Intramedullary nail design for Thais with novel multi-femur shape optimization approach

Received: 26 July 2024

Accepted: 2 January 2026

Published online: 18 January 2026

Cite this article as: Sayabut D., Virulsri C. & Tangpornprasert P. Intramedullary nail design for Thais with novel multi-femur shape optimization approach. *Sci Rep* (2026). <https://doi.org/10.1038/s41598-026-35172-7>

Duangkamon Sayabut, Chanyaphan Virulsri & Pairat Tangpornprasert

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

Intramedullary Nail Design for Thais with Novel Multi-femur Shape Optimization Approach

Duangkamon Sayabut^{1,2}, Chanyaphan Virulsri^{1,2,3,*}, Pairat Tangpornprasert^{1,2,3}

¹ Center of Excellence for Prosthetic and Orthopedic Implant, Chulalongkorn University, Bangkok 10330, Thailand

² Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

³ Biomedical Engineering Research Center, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

* Corresponding author E-mail: Chanyaphan.v@eng.chula.ac.th

ABSTRACT

Femoral fractures, often resulting from high-energy accidents, are commonly treated with intramedullary nails. Most currently used intramedullary nails are designed based on the anatomical characteristics of the Caucasian population, which are unsuitable for Asian populations due to differences in femoral medullary canal curvature. This mismatch causes stress in the femur, which leads to nail tip penetration. This research proposes a novel multi-femur shape optimization approach to design a long intramedullary nail that reduces stress on the femur by better conforming to the Thai femoral medullary canal. 60 CT scan images of Thai femurs were analyzed, and ten representative femurs were selected to cover the geometric diversity of the femur population using k-means clustering. The nail curve was modeled using four circular arcs, with radii optimized within a 700-1200 mm boundary. The nails were simultaneously optimized in the ten representative femurs utilizing a multi-objective genetic algorithm (MOGA) in Ansys, with the objective of minimizing the maximum principal stress in all ten femurs. The multi-femur shape optimized nail can reduce stresses in the representative femurs. Finite element analysis was conducted to compare the stresses induced by the multi-femur shape optimized nail with the stress caused by the commercial nails with radii of 1275 mm, 1500 mm, and 2000 mm. The results indicate that the multi-femur shape optimized nail significantly reduced the average maximum principal stress in randomly selected femurs. This research demonstrates that a multi-femur shape optimization approach can effectively design an intramedullary nail that reduces stress in a diverse set of femurs, making it a better fit for the Thai femur population. The optimized nail design was further evaluated in a verification model, including lag/distal screws and a one-leg-stance load; the stress in the multi-femur shape optimized nail was substantially lower than the commercial designs.

Keywords

Multi-objective shape optimization, Intramedullary nail design, Thai femur, Nail insertion simulation, femoral curvature

Introduction

Femoral fractures commonly result of high-energy trauma, such as falls or motor vehicle collisions. Treatment options include closed reduction, external fixation, and intramedullary devices. Among these, most subtrochanteric and intertrochanteric femoral fractures are treated surgically with intramedullary devices ¹, which is a specially designed rod inserted through the medullary canal to stabilize the fracture and promote healing. Compared with extramedullary fixation, intramedullary nailing offers several clinical advantages, including shorter hospital stays, faster recovery times, lower infection rates, and reduced re-operation rates ²⁻⁵.

Most commercially available intramedullary nails are designed based on the anatomical characteristics of the Caucasian population, with reported radii of curvature ranging from 1275 to 2000 mm ⁶⁻⁸. However, femoral morphology differs substantially between Asian and Caucasian populations. Chantarapanich et al. found that the average radius of the medullary canal in the Thai femur is 895 mm ⁹. Similarly, Cho et al. reported that the posterior cortex curvature of the Korean femur projected on the sagittal plane averages 814.43 mm ¹⁰. Su et al. measured 426 Chinese femurs, finding an average curvature radius of 971.44 mm ¹¹. Collectively, these findings indicate that Asian femurs generally exhibit smaller curvature radii than those assumed in current nail designs, suggesting that many existing nails are excessively straight for these populations.

Although some intramedullary nails have been redesigned to better accommodate Asian femoral anatomy, their radii of curvature remains relatively large ¹². This geometric mismatch can lead to complications such as anterior distal femur cortex penetration, particularly at the nail tip ¹³⁻¹⁶. Previous studies have shown that nails with more appropriate curvature can reduce

the risk of nail tip penetration^{13,14}. A mismatch between the nail curvature and the medullary canal is also likely to induce elevated stresses at the nail tip region, potentially causing local femoral damage. Therefore, intramedullary nail curvature should be designed with explicit consideration of stress distribution within the femur during insertion.

Several studies have proposed alternative approaches to characterize medullary canal geometry for improved nail design. Akman et al. modeled the medullary cavity as a chord of ring cyclide and measured anterior and posterior border radii of 1478 mm and 787 mm, respectively. They reported an optimal nail curvature radius between 900 and 990 mm¹⁷. Ren et al. described femoral curvature using a tangent-based parameter incorporating femoral length and radius of curvature¹⁸. Arnone et al. generated a surface-averaged femur model to represent a population-based femoral geometry and fitted a spline along the medullary canal to derive a curvature function for nail design. Finite element analyses showed that this design reduced von Mises stress in the femur compared with conventional nails with 2 m and 3 m radii¹⁹. However, none of these studies considered femoral stress as a primary design criterion. Moreover, excluding Arnone et al., most studies assumed that the medullary canal follows a circular arc^{9-11,20}.

Recent anatomical studies suggest that femoral curvature is non-uniform along its length. Abdelaal et al. found that femoral curvature varies between the proximal, middle, and distal regions. Japanese femurs were found to resemble a parenthesis-shaped curve²¹, whereas Chinese femurs exhibited a J-shaped curvature²². These findings imply that non-circular curvature profiles may better conform to femoral anatomy and more effectively minimize stress concentrations.

Shape optimization has been widely used in the design of orthopedic implants, including cementless hip prostheses²³ and femoral fixation devices²⁴. However, most existing studies perform optimization using a single femur model, which may limit the applicability of the resulting design to a broader population.

This research aims to propose a novel multi-femur shape optimization approach to design long intramedullary nails that reduce stress in Thai femurs during insertion. The proposed method focuses on minimizing insertion-induced residual stresses arising from geometric mismatch between the nail and the medullary canal. In this approach, a diverse set of representative femurs will undergo simultaneous shape optimization with the same nail design. These representative femurs were chosen from a dataset of 60 Thai femurs, which were classified into ten groups using k-means clustering, with one representative femur selected from each group. To evaluate the robustness of the optimized design, additional verification under physiological loading conditions was conducted.

Materials and methods

Femoral specimen

Sixty computed tomography (CT) scans of Thai femurs (39 females, 21 males; mean age 56.27 years old, range 17-89 years) were retrospectively obtained from King Chulalongkorn Memorial Hospital, Bangkok, Thailand, as part of a previous study. All images were collected with informed consent.

The use of these de-identified CT data for the present computer-based simulation study was reviewed and granted an exemption from ethical approval by the Institutional Review Board of the Faculty of Medicine, Chulalongkorn University (COE No. 099/2025, IRB No. 0940/68). The study involved no direct human participants and contained no identifiable personal information. All procedures were conducted in accordance with relevant guidelines and regulations, as well as the principles of the Declaration of Helsinki.

Design specification

A long cephalomedullary nail was considered in this study. The nail is inserted into the medullary canal through the greater trochanter. The design consists of two segments: a proximal segment and a distal segment. The geometry of the proximal part was defined based on commercially available nail designs. The proximal cross-sectional diameter was set to 15 mm, and the proximal lateralization angle was fixed at 4.5°^{6-8,12,25}. The distal cross-sectional diameter was set to 8 mm, selected to accommodate the average femoral shaft isthmus diameter of Thai femurs (9.94 ± 1.76 mm²⁶). The total nail length was 320 mm, with a distal segment length of 230 mm. This length was chosen to ensure compatibility with all specimens, as the minimum femoral shaft length measured in the dataset was 234.23 mm. The nail was assumed to be manufactured from a titanium alloy. A schematic of the nail geometry is shown in Figure 4.

Conceptual design

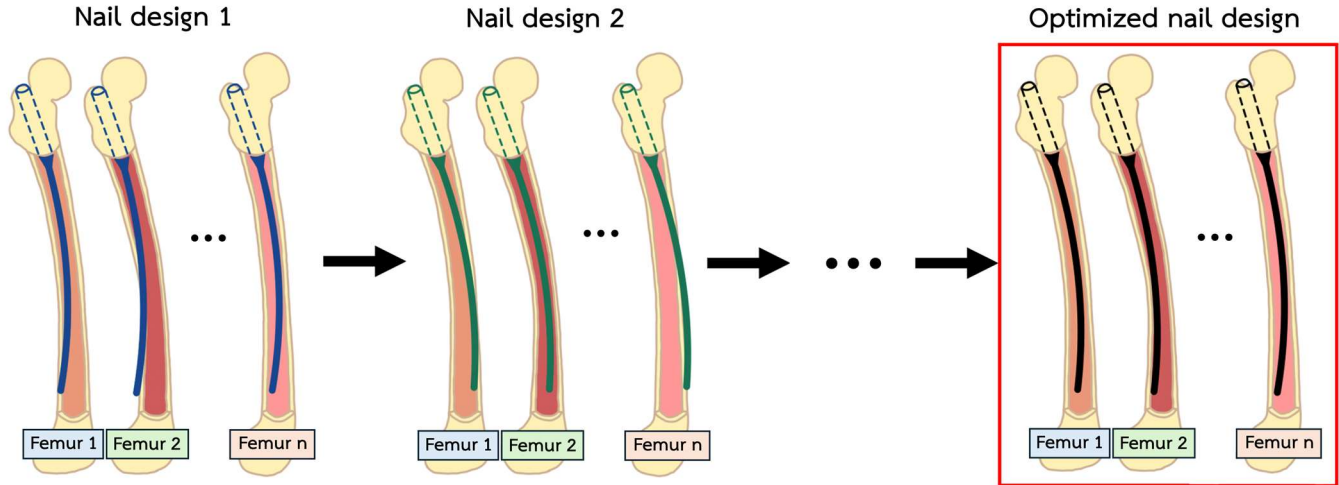


Figure 1. Illustration of multi-femur shape optimization, where a nail is simultaneously optimized in a set of n femurs to achieve the best nail design (optimized nail) that minimizes stresses in all n femurs

The nail was designed using a shape optimization method, which is commonly applied in the development of orthopedic fixation devices. In conventional approaches, shape optimization is typically performed using a single femur model. Consequently, the resulting optimized design may be well suited to that specific femur but may not perform optimally for other femoral geometries, even when a representative or average femur is used.

In this research, a novel multi-femur shape optimization approach is proposed. In this approach, the nail is simultaneously optimized using the entire selected representative femurs, as shown in Figure 1. These representative femurs were chosen to capture the geometric variability of the femur population. The resulting design, hereafter referred to as the multi-femur shape-optimized nail, is therefore expected to provide improved geometric conformity and reduced stress across the femur population.

Clustering and selecting representative femurs

To reduce computational resource requirements, the optimization process was performed using ten femur models. The original dataset of 60 femoral specimens was therefore classified into ten clusters, and one representative femur was selected from each cluster for use in the multi-femur shape optimization.

In this study, the term representative femur refers to a femoral geometry selected via clustering to capture population-level anatomical variability and subsequently used during the multi-femur shape optimization process.

CT scans of 60 Thai femurs in DICOM format were imported into MATLAB (R2020a). Each femur was reconstructed as a three-dimensional voxel-based point cloud. Cortical bone was segmented using the `imsegkmeans3()` function, which applies a k-means clustering-based volume segmentation algorithm. The models were then cropped to include only the femoral shaft, as all measurements were restricted to this region. The resulting femoral shaft geometry resembled a curved hollow structure, with the medullary canal forming the internal cavity (Figure 2a). A Boolean operation was subsequently performed to isolate the medullary canal model (Figure 2b).

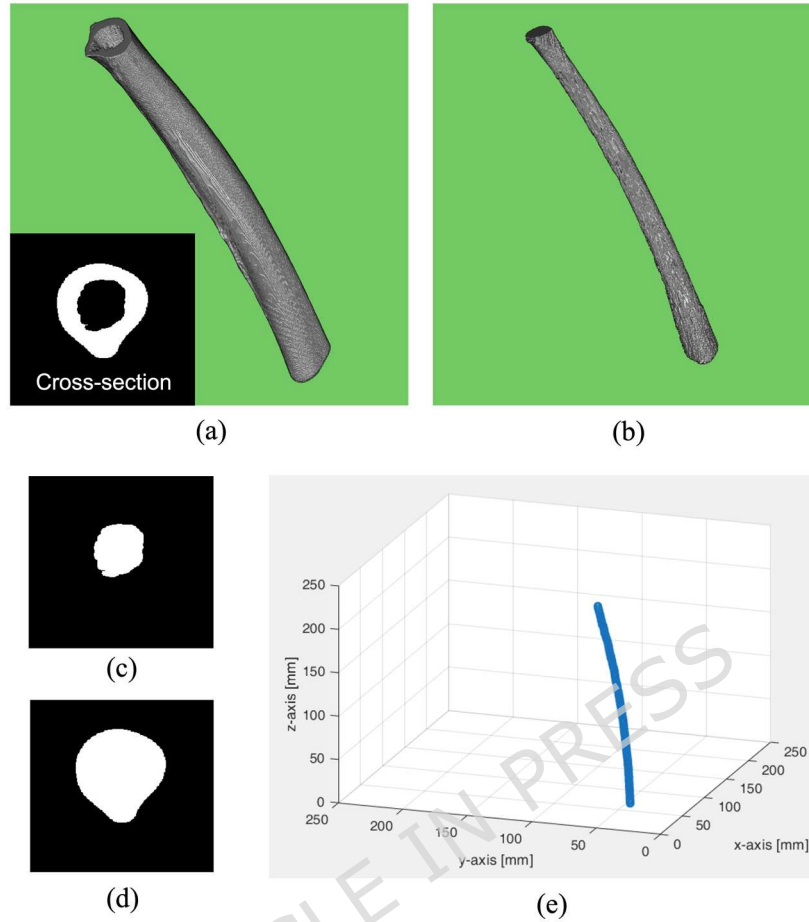


Figure 2. (a) 3D model and cross-sectional area of the cortical femoral shaft (b) 3D model of the medullary canal (c) Cross-sectional area of the medullary canal (d) Cross-sectional area of the whole femur (e) Centroid of each cross-section slide of the medullary canal

Three geometric parameters were extracted from each femur for clustering: (1) the femoral radius of curvature, (2) the cross-sectional area ratio at the level of the lesser trochanter, where the diameter of the designed nail changes, and (3) the cross-sectional area ratio at the length of 230 mm from the lesser trochanter, corresponding to the distal end of the designed nail. The radius of curvature characterizes the geometric conformity between the nail centerline and the femoral medullary canal. The cross-sectional area ratios at these two locations quantify cortical bone thickness, which is clinically relevant to bone strength and fracture risk. These parameters were selected because they directly influence stress distribution within the femur during nail insertion.

To determine the radius of curvature, the centroids of the medullary canal cross-sections were calculated along the femoral shaft from the lesser trochanter to a point 230 mm distally (Figure 2e). A circular arc was then fitted to these centroids, and the corresponding radius was defined as the femoral radius of curvature. For the cross-sectional area ratio, the ratio of the total femoral cross-sectional area (including cortical bone and medullary canal; Figure 2d) to the medullary canal area alone (Figure 2c) was calculated to describe cortical bone thickness. These measurements were taken at both the lesser trochanter and the 230 mm length from the lesser trochanter location.

Based on these three parameters, the 60 Thai femurs were classified into ten clusters using the k-means clustering algorithm. K-means was selected because it efficiently partitions data into a predefined number of clusters with approximately spherical distributions and provides cluster centroids that serve as representative values for each group. By minimizing within-cluster variance, the algorithm ensures that femurs within the same cluster are similar across all selected dimensions^{27,28}. The clustering quality was evaluated using the silhouette coefficient. For $k = 10$, the mean silhouette coefficient was 0.4379, indicating moderate and acceptable separation among clusters. The results of the clustering analysis are shown in Figure 3. One representative femur was selected from each cluster by identifying the femur with the smallest Euclidean distance to the corresponding cluster centroid.

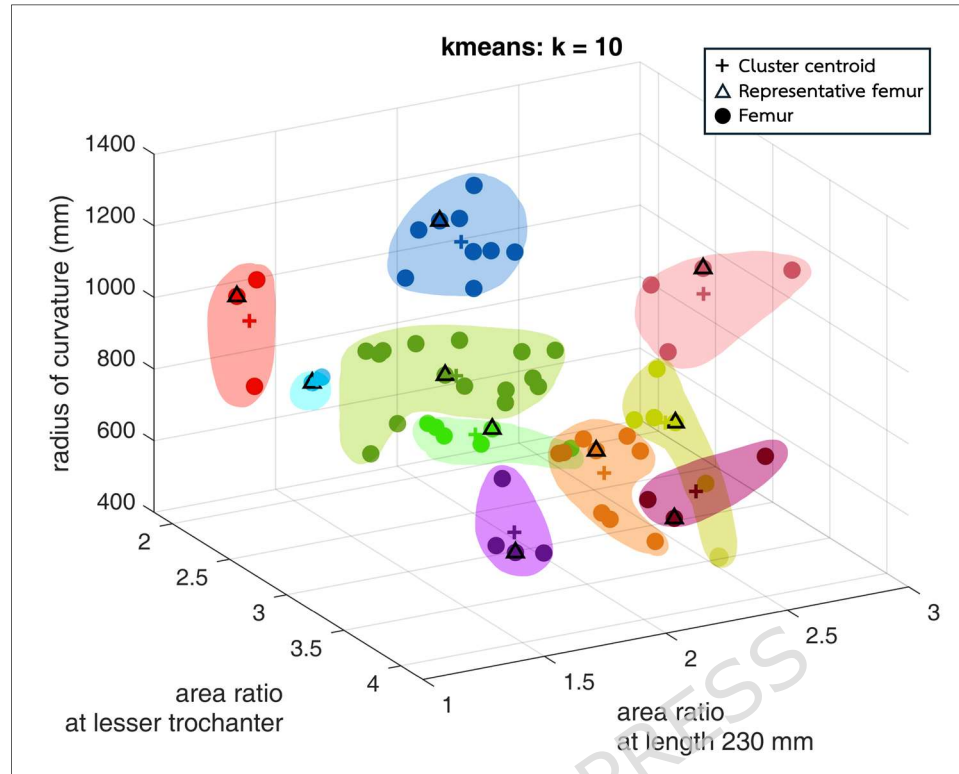


Figure 3. k-means clustering of 60 Thai femurs based on three geometric parameters: radius of curvature (mm), cross-sectional area ratios at the lesser trochanter (dimensionless), and cross-sectional area ratio at 230 mm from the lesser trochanter (dimensionless)

3D femur modeling

Three-dimensional (3D) reconstruction models of the ten representative femurs were generated from CT scan data. Each femur model consisted of two regions: the proximal femur and the femoral shaft. The proximal femur was assumed to consist entirely of cortical bone, whereas cortical bone in the femoral shaft was segmented using a minimum Hounsfield unit (HU) threshold of 662²⁹. The proximal and shaft models were subsequently merged to form a single solid femur model.

Multi-femur shape optimization

Multi-femur shape optimization overview

The nail geometry was optimized using the Direct Optimization module in Ansys (version 2021 R2) to minimize stress in the femurs. In each optimization iteration, a design point, defined as a specific set of nail centerline curvature design variables, was generated and used to model the intramedullary nail inserted into each representative femur. The nail insertion process was then simulated for all femurs, and the resulting maximum stress values were extracted as the optimization responses.

Consistent with the objective of designing a single intramedullary nail suitable for multiple femoral geometries, each design point represented one unique nail geometry that was evaluated across all representative femurs within a given optimization iteration. For each design point, the same nail geometry was inserted into every representative femur, and the finite element modeling procedure was kept identical in terms of nail dimensions, mesh settings, material properties, and insertion conditions.

Intramedullary nail design parameters

Because the femoral medullary canal does not follow a single circular arc^{21,22}, the curvature of the nail distal segment was represented using four consecutive circular arcs. The radii of these arcs, assigned as r_1 , r_2 , r_3 , and r_4 , are shown in Figure 4. These four radii were defined as the design variables and collectively constitute a single design point in the optimization process.

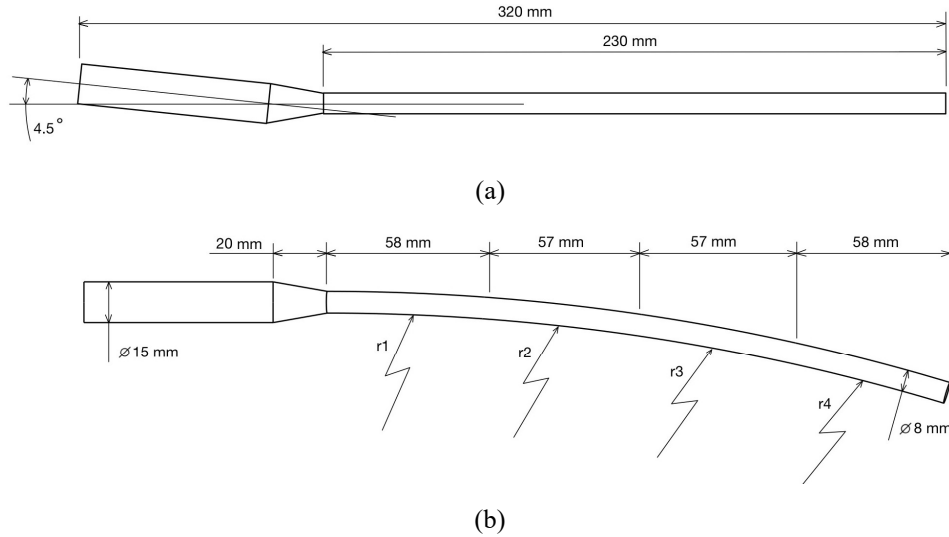
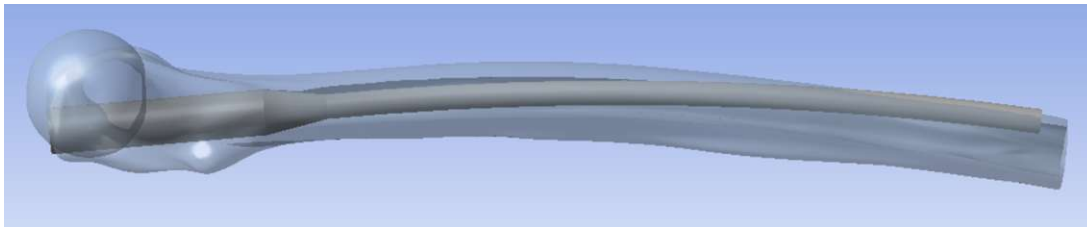


Figure 4. A nail drawing (a) coronal plane view (b) sagittal plane view

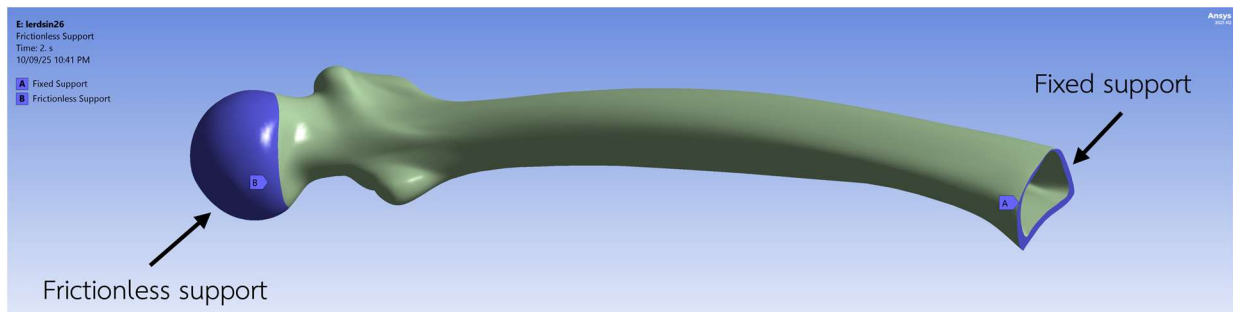
Finite element model

The 3D STEP (STP) models of the ten representative femurs were imported into Ansys DesignModeler. As described previously, the intramedullary nail corresponding to each design point was first modeled in DesignModeler and then inserted into each representative femur. To prepare the femur model, a cylindrical hole with a diameter of 15 mm was created through the greater trochanter, matching the proximal diameter of the nail. The alignment of the proximal nail segment within the prepared proximal femur is shown in Figure 5a.

The assembled femur–nail models were subsequently transferred to ANSYS Mechanical for finite element analysis. The nail was assigned titanium alloy material properties, while the femur material properties were defined as listed in Table 1³⁰. These material properties were held constant throughout the multi-femur shape optimization to ensure computational consistency across all design points. All finite element models were discretized using three-dimensional solid elements. A mesh sensitivity analysis was performed by progressively refining the mesh until changes in peak stress values was less than 1%, after which further refinement produced negligible changes in the results. The nail components were meshed with a uniform element size of 2.2 mm, while the femoral models were meshed with an element size of 5 mm. The same meshing strategy was applied consistently across all simulations. A frictionless support was applied to the femoral head, and fixed support was applied to the distal end of the femur at a cut surface above the condyle, as shown in Figure 5b. The contact between the proximal segment of the nail and the prepared hole in the proximal femur was defined as bonded (Figure 5c).



(a)



(b)

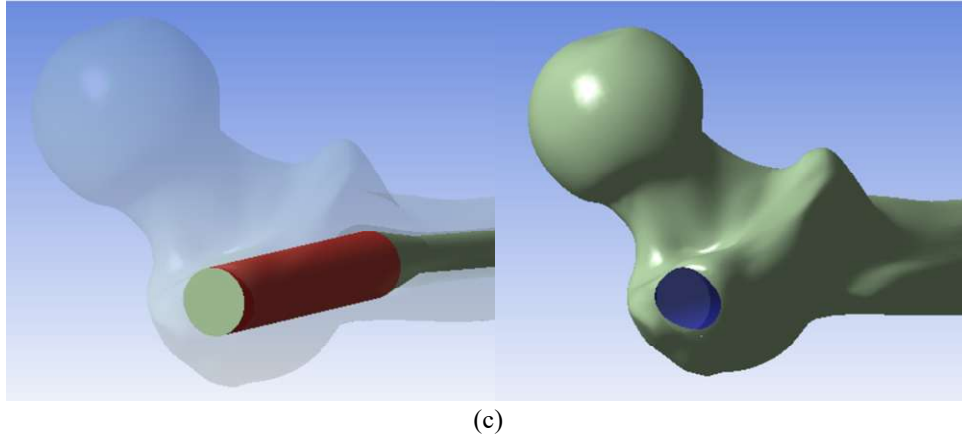
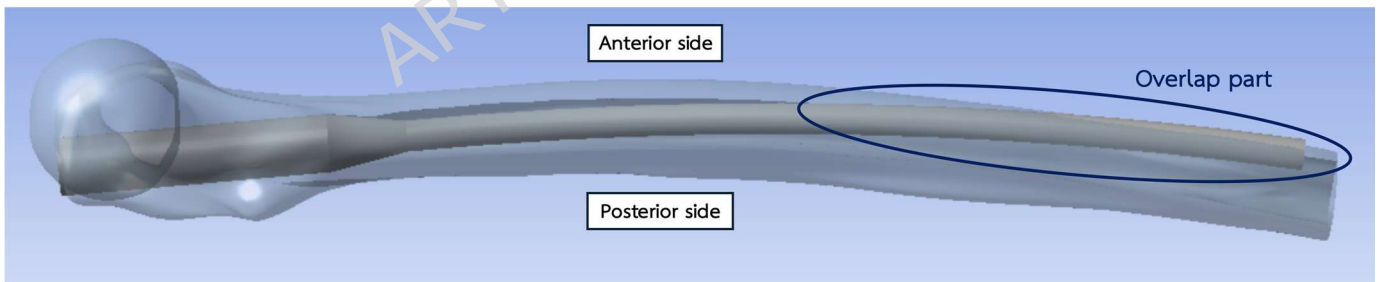


Figure 5. (a) An alignment set-up of nail and femur (b) frictionless support at the femoral head and fixed support at the cutting area above the condyle (c) Bonded contact between the proximal part

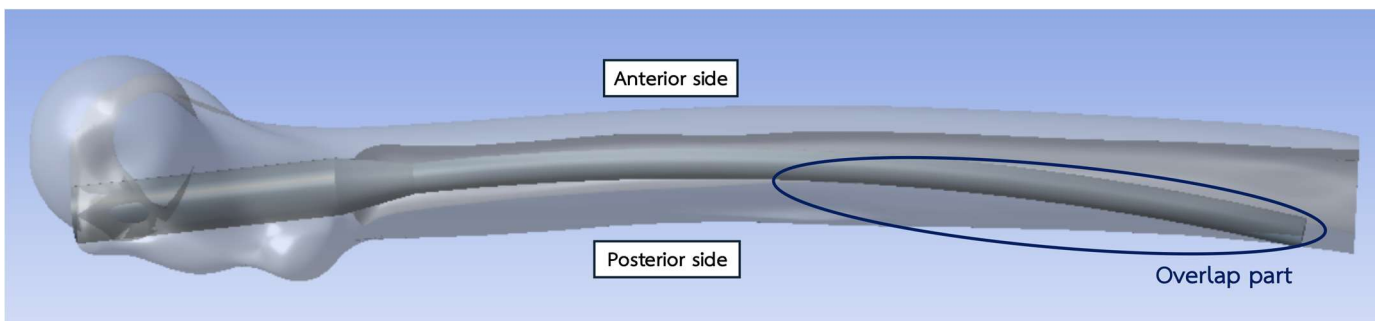
During nail insertion, geometric mismatch between the nail centerline curvature and the medullary canal may cause local overlap between the nail and cortical bone. When the nail curvature is too straight, overlap may occur on the anterior cortex (Figure 6a), whereas an excessively curved nail may cause overlap on the posterior cortex (Figure 6b). To model the insertion of the distal nail segment into the medullary canal without artificial penetration, the inner surface of the canal was divided into anterior and posterior contact regions. The insertion process was simulated using three sequential static analysis steps, as illustrated in Figure 6c.

In Step 1, no contact was defined between the nail and the canal surfaces, and a force was applied to the distal end of the nail in the posterior direction to move it away from the anterior cortex. In Step 2, contact was activated between the nail and the anterior canal surface, and a force was applied in the opposite direction to bring the nail into contact with the anterior surface while preventing posterior overlap. In Step 3, contact was activated on both the anterior and posterior canal surfaces, and the applied force was removed, allowing the nail to elastically recover toward its original shape and induce stress within the femur.

The maximum principal stress within the femur was selected as the output parameter for the optimization process and was evaluated for all representative femurs at each design point.



(a)



(b)

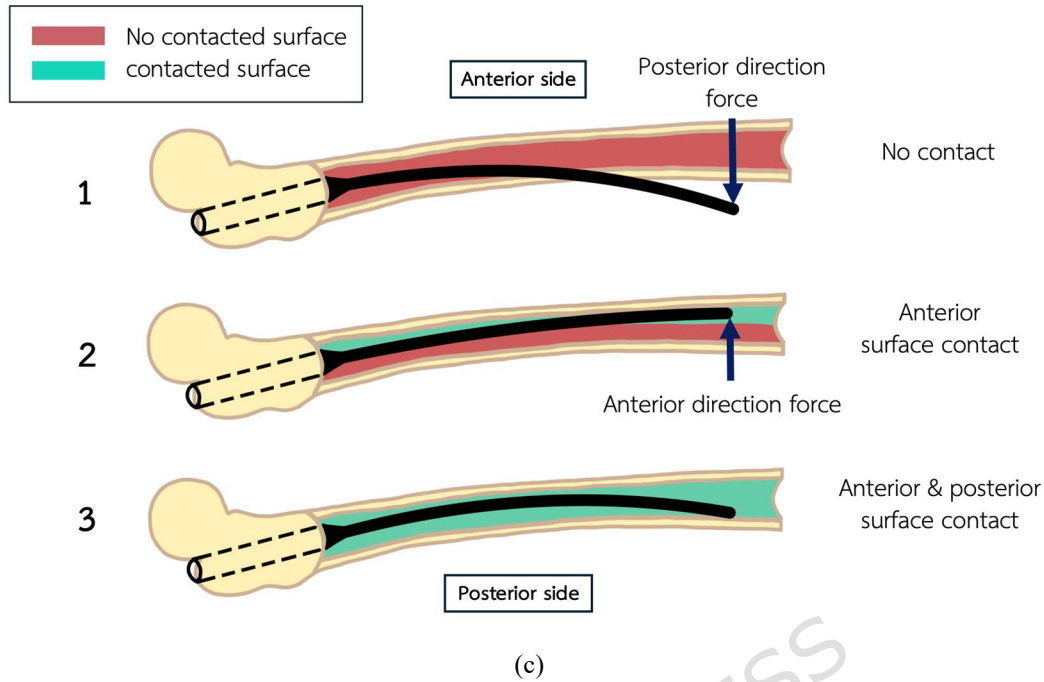


Figure 6. (a) Overlap of the distal part of the nail on the anterior side femur (b) Overlap of the distal part of the nail on the posterior side femur (c) Illustration of the nail insertion process, which is divided into three steps: 1. Force is applied to the nail's distal end in the posterior direction to move it away from the anterior surface 2. Contact is set with the anterior surface, and force is applied in the opposite direction 3. Contact is set with both anterior and posterior surfaces, and the force is released, allowing the nail to spring back and ensuring it is enclosed within the femoral canal.

Material	Type	Mechanical properties
Titanium Alloy	Linear isotropic	$E = 96 \text{ GPa}$, $\nu = 0.36$
Ti-6Al-4V	Linear isotropic	$E = 113.8 \text{ GPa}$, $\nu = 0.342$
Cortical bone	Orthotropic	$E_x = 12 \text{ GPa}$, $E_y = 13.4 \text{ GPa}$, $E_z = 20 \text{ GPa}$ $G_{xy} = 4.5 \text{ GPa}$, $G_{yz} = 6.2 \text{ GPa}$, $G_{xz} = 5.6 \text{ GPa}$ $\nu_{xy} = 0.38$, $\nu_{yz} = 0.23$, $\nu_{xz} = 0.22$

Table 1. Material properties^{30,31}

Shape optimization process

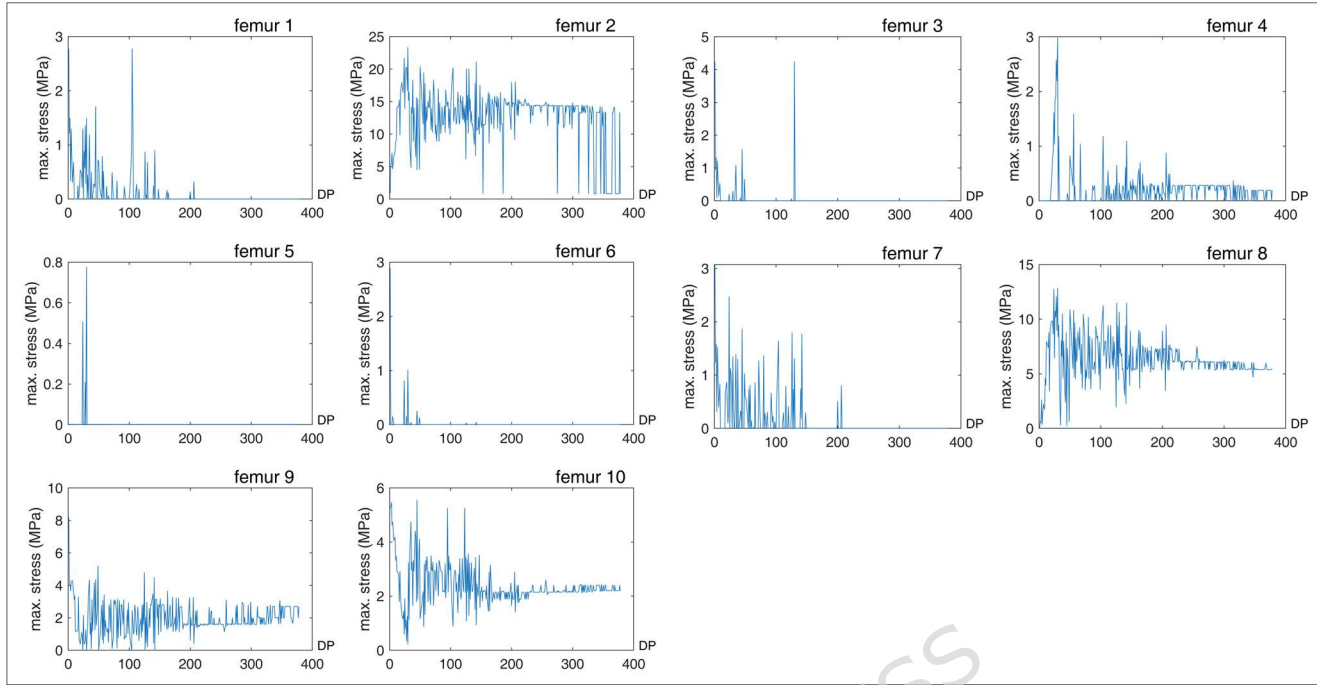
The nail geometry was optimized using the Direct optimization module in Ansys with the Multi-Objective Genetic Algorithm (MOGA) method. Throughout the optimization, a single nail geometry was evaluated across all ten representative femurs, such that all femurs shared the same nail centerline curvature at each design point. The design variables were the radii of the four circular arcs (r_1 , r_2 , r_3 , and r_4), which were constrained within a range of 700–1200 mm.

The multi-objective optimization aimed to minimize the maximum principal stress in each representative femur simultaneously. An initial population of 30 design points was generated, and 30 new design points were produced in each subsequent iteration using the MOGA framework. The optimization process was terminated when either the maximum allowable Pareto percentage reached 70% or the convergence stability percentage reached 2%.

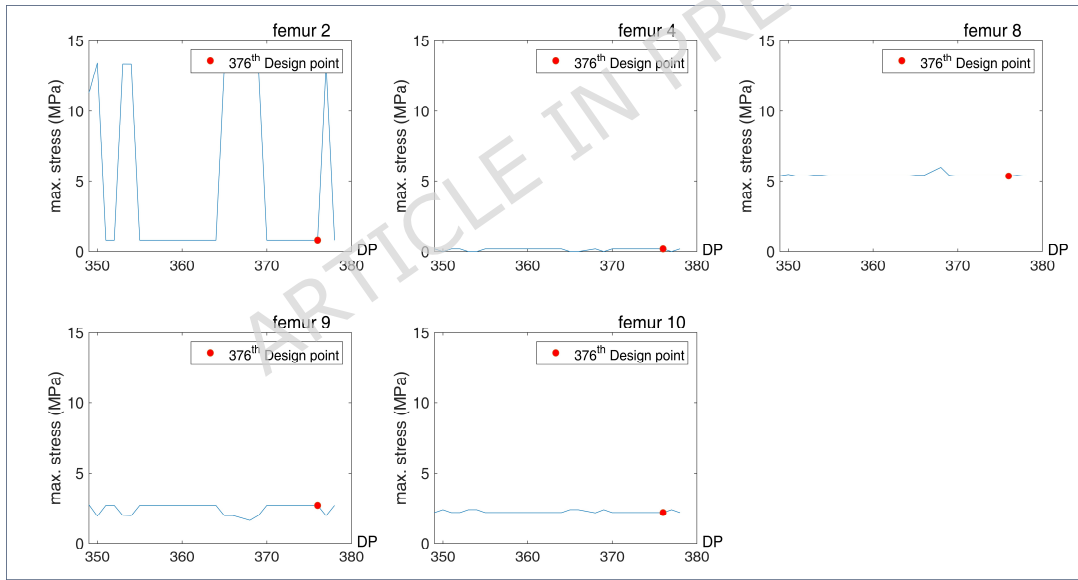
Results and discussion

The multi-femur shape-optimization results

The optimization process terminated at the 14th iteration, when the Pareto percentage reached 70%. The maximum principal stresses obtained for each representative femur at every design point (DP) are presented in Figure 7a. Among the final 30 design points, representative femurs 1, 3, 5, 6, and 7 exhibited convergences toward stress values close to zero, indicating that the optimization objective was effectively achieved for these femoral geometries. In contrast, although stresses in representative femurs 2, 4, 8, 9, and 10 were substantially reduced, they did not converge to zero. Consequently, selection of the final design point was governed by the stress responses of these femurs.



(a)



(b)

Figure 7. (a) Maximum principal stresses (MPa) of 10 representative femurs occurred at each design point (DP) during the nail optimization process (b) The maximum principal stresses (MPa) occurred in femurs 2, 4, 8, 9, and 10 among the last 30 design points

Figure 7b illustrates the maximum principal stresses for representative femurs 2, 4, 8, 9, and 10 across the final 30 design points. The maximum stress values among these femurs ranged from approximately 5.38 to 14.19 MPa. Design point 376 (DP 376; red markers in Figure 7b) was selected as the final optimized design because it yielded one of the lowest maximum stress values (≈ 5.38 MPa) across all representative femurs. This selection follows a min-max criterion³², ensuring that the worst-case stress condition among the femurs was minimized. The geometric parameters defining the multi-femur shape-optimized nail are summarized in Table 2.

r1 (mm)	r2 (mm)	r3 (mm)	r4 (mm)
1060	755	800	1195

Table 2. Details of the multi-femur shape-optimized nail curve

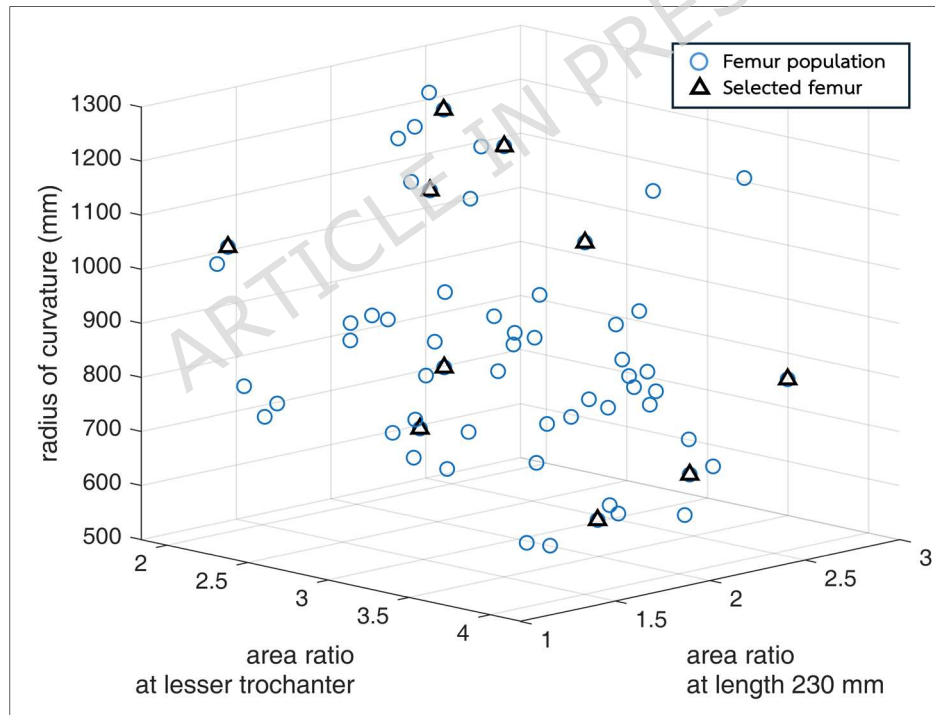
Comparison with the commercial nails

To evaluate the robustness of the optimized nail geometry under more clinically representative conditions, additional finite element simulations were performed. In this verification analysis, the nail material was updated to Ti-6Al-4V with an elastic modulus of 113.8 GPa and a Poisson's ratio of 0.342³¹ (Table 1), reflecting commonly reported properties of clinical implants.

For comparison of insertion-induced femoral stress, an independent set of ten femurs was selected from the original dataset of 60 femurs. These femurs were distinct from the representative femurs used during the shape optimization process and were used exclusively for comparative evaluation. The sample size of ten femurs was selected to balance computational feasibility with anatomical representativeness. Rather than being randomly chosen, the femurs were deliberately selected to span the geometric variability of the population based on three preliminary parameters: femoral radius of curvature, cross-sectional area ratio at the lesser trochanter, and cross-sectional area ratio at a length of 230 mm distal to the lesser trochanter.

This selection strategy is consistent with the shape-optimization framework adopted in this study, in which representative femurs were identified via K-means clustering to capture population-level geometric variability. In both cases, the emphasis is placed on geometric coverage rather than statistical population inference. Accordingly, a sample size of ten femurs was considered sufficient to provide a robust and computationally tractable comparison of insertion-induced stress between the multi-femur shape-optimized nail and commercial nail designs.

The selected femurs are shown in Figure 8a. Three commercial nails with radii of 1275, 1500, and 2000 mm were used for comparison⁶⁻⁸. Both the multi-femur shape-optimized nail and the commercial nails were inserted into those selected femurs, and the resulting maximum principal stress induced during insertion was evaluated (Figure 8b).



(a)

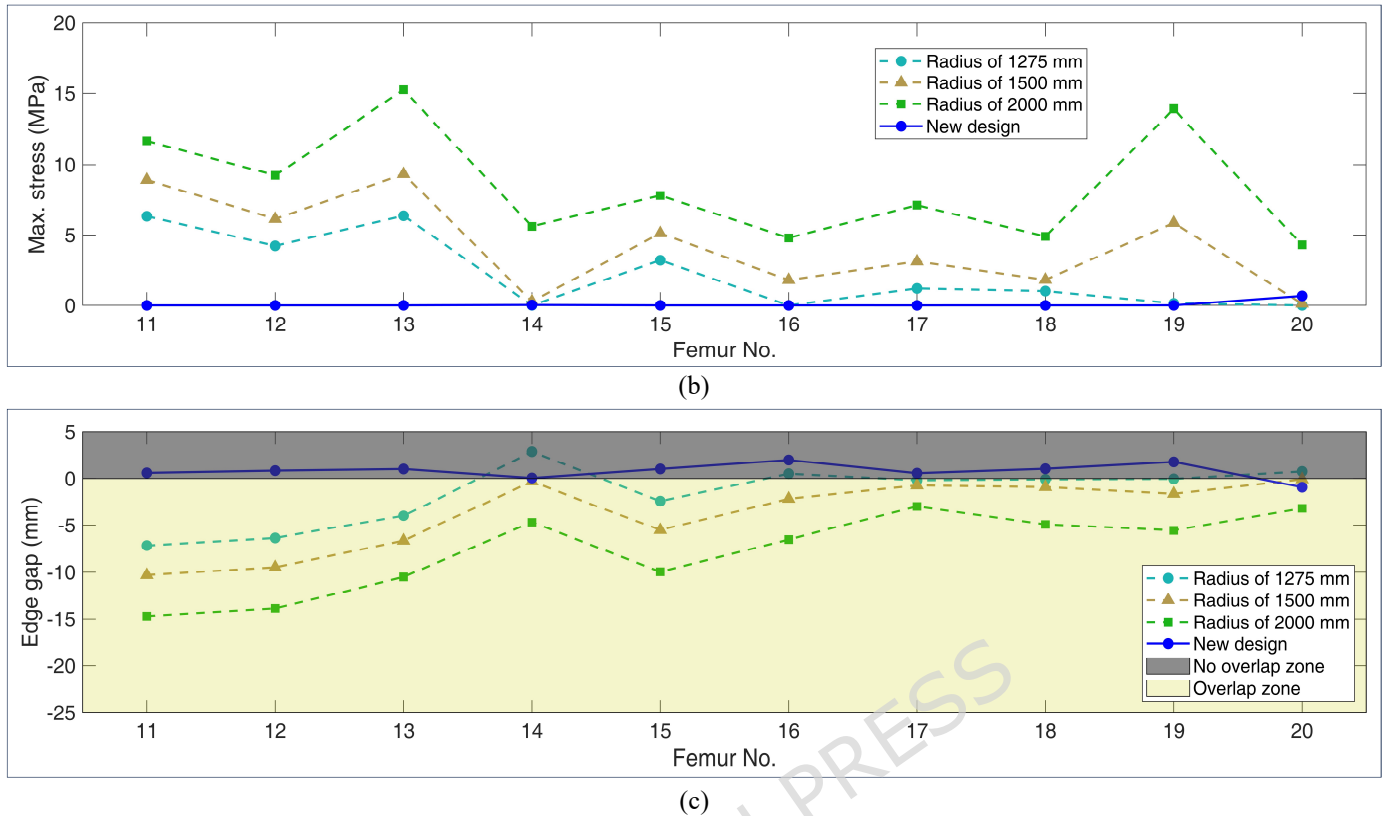


Figure 8. (a) Selected femurs from 60 femur population for comparing the new design with an average radius nail (b) The maximum principal stresses (MPa) of the selected femurs (c) The gap (mm) between the edges of the femur canal and the nail (positive values = gap, negative values = overlap)

Figure 8b shows that insertion-induced stresses were consistently higher when commercial nails were used compared with the multi-femur shape-optimized nail. Insertion of the optimized nail resulted in stress values approaching negligible levels, whereas all commercial nails produced substantially higher stresses. These differences were statistically significant based on the Wilcoxon signed-rank test, with median stress differences and corresponding 95% confidence intervals indicating consistently higher stresses for commercial nails ($p = 0.036, 0.009, \text{ and } 0.005$ for radii of 1275, 1500, and 2000 mm, respectively).

Across the ten comparison femurs, the mean maximum principal stress was 0.0035 ± 0.0058 MPa for the multi-femur shape optimized nail, compared with 2.54 ± 2.89 MPa for the 1275 mm nail, 4.93 ± 4.09 MPa for the 1500 mm nail, and 10.08 ± 4.66 MPa for the 2000 mm nail. These correspond to stress reductions of 99.86%, 99.93%, and 99.96%, respectively, relative to the commercial designs. Although the percentage reductions are numerically large, the absolute stress values confirm that the optimized nail substantially minimizes insertion-induced stress.

During the simulated nail insertion process, external forces were applied to prevent artificial overlap between the nail and the cortical bone. Such overlap causes contact forces that induce stress within the nail, which is subsequently transferred to the femur. Therefore, femoral stress is directly related to the degree of geometric overlap between the nail and the medullary canal. In Figure 8c, positive values (gray region) represent the minimum clearance between the nail and the femur, whereas negative values (yellow region) indicate overlap. The results demonstrate a clear correlation between overlap magnitude and femoral stress, with greater overlap producing higher stress levels. This further confirms that the multi-femur shape-optimized nail minimizes femoral stress by better conforming to the natural curvature of the medullary canal.

Comparison with existing studies

The importance of implant geometry in influencing bone stress distribution has also been highlighted in other biomechanical investigations, where fixation constructs with different configurations demonstrated distinct load-transfer characteristics under physiological conditions³³. Although those studies focused on distal femoral fixation, the underlying principle similarly supports our finding that improving geometric conformity between the nail and medullary canal can reduce stress concentrations and enhance mechanical compatibility.

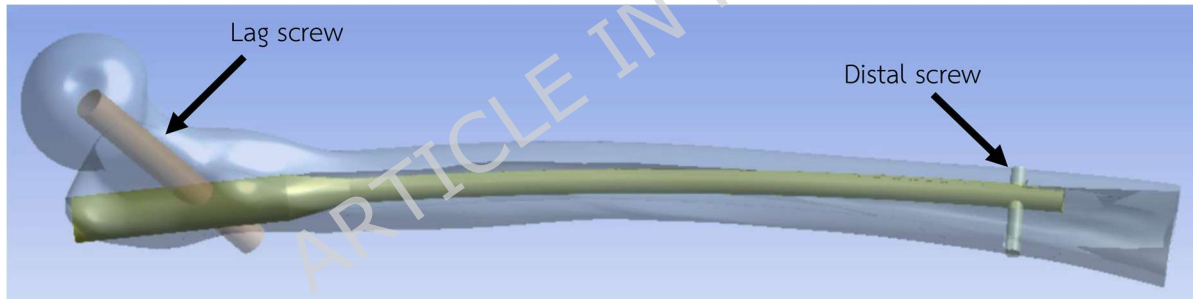
Verification under physiological loading conditions

To evaluate the mechanical robustness of the optimized nail under clinically relevant conditions, an additional finite element model was developed using femur no. 15 and four nail designs: the multi-femur shape-optimized nail and three commercial nails with radii of curvature of 1275, 1500, and 2000 mm. Lag screws and distal locking screws were included in the model, as illustrated in Figure 9a. Screw-bone and screw-nail interfaces were defined as bonded to represent rigid fixation, while the nail–femur interface was modeled as frictionless to simulate post-reaming clearance ³⁴.

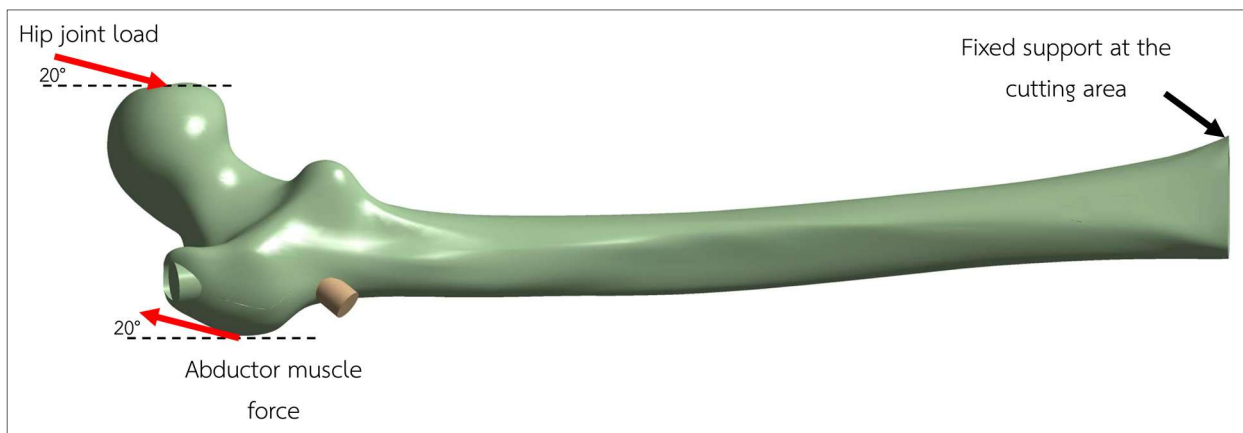
Physiological loading corresponding to a one-leg-stance condition was applied to the model to represent normal ambulation. A peak hip joint load of 2058 N (approximately 300% body weight) was applied at the femoral head, together with an abductor muscle force of 686 N (approximately 100% body weight) acting laterally. Both loads were applied at an angle of 20° from the vertical in the frontal plane ³¹. Fixed support was applied at the distal end of the femur, as shown in Figure 9b. Material properties for Ti-6Al-4V and cortical bone were identical to those used in the insertion simulations (Table 1) to ensure consistency.

The resulting stress distributions are presented in Figure 9c and summarized in Table 3. The optimized nail exhibited substantially lower stress levels than the commercial designs. Specifically, the maximum stress around the distal screw hole was 3.67 MPa for the optimized nail, compared with 91.68, 180.62, and 203.26 MPa for the commercial nails with radii of 1275, 1500, and 2000 mm, respectively. Similarly, stress in the transition region between the proximal and distal portions of the nail was 23.84 MPa for the optimized design, whereas stresses of 82.73, 142.83, and 254.78 MPa were observed for the commercial nails.

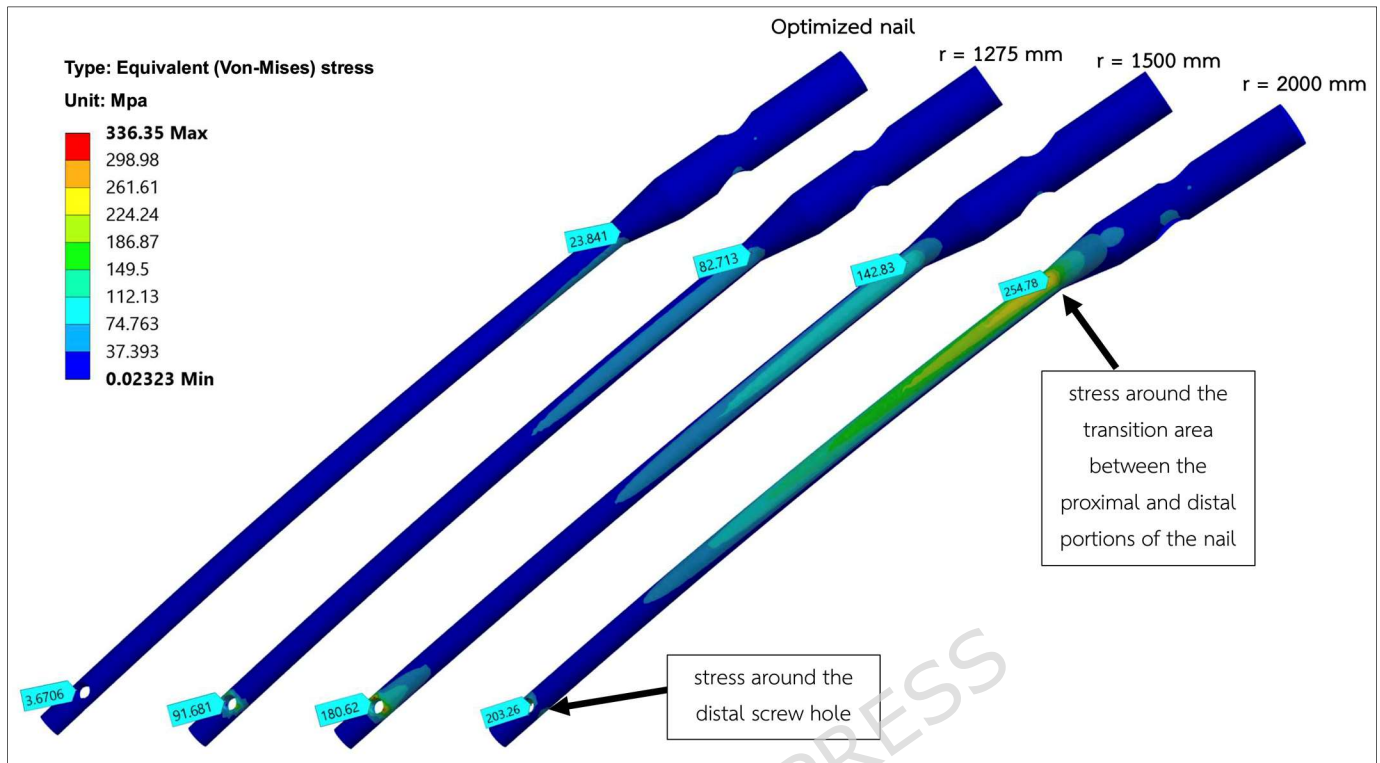
The improved conformity of the optimized nail to the femoral medullary canal promotes a more uniform stress distribution within the implant, thereby enhancing mechanical stability under physiological loading. This increased conformity may also improve resistance to bending and torsional loads during fracture healing ³⁵. Overall, these results demonstrate a substantial reduction in implant stress and suggest a lower risk of fatigue failure compared with conventional nail designs.



(a)



(b)



(c)

Figure 9. (a) the verification simulation set-up with lag screw and distal screw (b) a one-leg-stance loading at the femoral head and fixed support at the distal cutting area (c) the verification results of all four nails showing Von-Mises stress (MPa) around the distal screw hole and around the transition area between the proximal and distal portions of the nail

	The multi-femur shape-optimized nail	Commercial nails		
		r = 1275 mm	r = 1500 mm	r = 2000 mm
Von-Mises stress around the distal screw hole (MPa)	3.67	91.68	180.62	203.26
Von-Mises stress around the transition area (MPa)	23.84	82.73	142.83	254.78

Table 3. The verification simulation results, showing Von-Mises stress (MPa) around the distal screw hole and around the transition area between the proximal and distal portions of the nail

Clinical relevance

From a clinical standpoint, excessive stress concentration within intramedullary nails, especially around distal screw holes and geometric transition regions, has been associated with implant fatigue, intraoperative mismatch, and an increased risk of cortical perforation. The comparative simulations in this study demonstrate that the multi-femur shape-optimized nail substantially reduces implant stress under physiological loading conditions compared with commonly used commercial designs. These findings suggest that improved geometric conformity between the nail and the femoral medullary canal can mitigate stress concentrations that are clinically implicated in implant-related complications.

Importantly, reduced stress near the distal nail tip and curvature transition regions is clinically relevant for lowering the risk of anterior cortical perforation, particularly in femurs with pronounced sagittal curvature. Such complications have been reported when relatively straight nails are inserted into more curved medullary canals¹³⁻¹⁶, leading to localized cortical overload during insertion or postoperative loading. By more closely following the native femoral curvature, the optimized nail may reduce unintended contact forces between the implant and cortical bone, thereby decreasing the likelihood of cortical breach.

From a surgical perspective, improved conformity between the nail centerline curvature and the medullary canal may facilitate a smoother insertion process. A nail that better matches the natural femoral curvature is expected to advance more easily along the canal, potentially reducing resistance during insertion and decreasing the need for excessive intraoperative force or repeated reaming. This may improve tactile feedback for the surgeon, reduce operative difficulty, and lower the risk of iatrogenic femoral damage during nail advancement, particularly in anatomically curved femurs.

Furthermore, the proposed multi-femur shape-optimization framework supports a population-informed approach to implant design. By accounting for anatomical variability across a femoral population rather than optimizing for a single femur geometry, this strategy may reduce the likelihood of intraoperative mismatch between the implant and patient anatomy. Clinically, this could translate into more consistent implant performance across patients, reduced reliance on intraoperative adjustments in nail selection or trajectory, and improved reproducibility of surgical outcomes.

Although the present findings are based on finite element simulations, the observed differences in implant stress under physiologically relevant loading conditions highlight the mechanical sensitivity of nail performance to geometric conformity. These results motivate further experimental and cadaveric investigations to assess the mechanical behavior of the optimized design under more realistic conditions.

Future industrial implementation

Beyond its clinical implications, the proposed multi-femur shape-optimization framework offers a pathway for domestic implant development and manufacturing. By incorporating population-specific femoral geometry into the design process, this approach enables the development of intramedullary nails tailored to regional anatomical characteristics. For countries such as Thailand, where most commercially available nails are imported and primarily designed based on Western anatomical data, population-informed designs present an opportunity to improve implant conformity while reducing reliance on imported devices.

Limitations and future works

Several limitations of this study should be noted. First, the proposed optimization framework was evaluated using finite element simulations, and no experimental or cadaveric validation was performed. Such validation will be necessary to confirm the biomechanical behavior of the optimized nail under more realistic conditions.

Second, lag and distal locking screws were excluded during the shape optimization process to reduce computational complexity. Although verification analyses incorporating screw fixation under physiologically relevant loading conditions demonstrated reduced implant stresses, future work should include screws directly within the optimization loop to more accurately represent postoperative conditions.

Third, the femoral geometries used in this study were derived from a Thai population, which may introduce population-specific anatomical bias and limit direct generalization to other ethnic groups. Expanding the dataset to include diverse populations would improve the broader applicability of the proposed approach.

Fourth, only static loading conditions were considered during optimization and verification. Dynamic loading scenarios, such as gait-related or impact forces, were not included and may influence implant stress distribution and long-term performance. Incorporating such loading conditions would provide a more comprehensive assessment of implant behavior.

Finally, the internal bone structure was simplified by omitting the cancellous bone region due to segmentation and meshing constraints, and homogeneous material properties were assumed. Additionally, the optimization objective was limited to maximum principal stress. Future studies may benefit from incorporating more detailed bone material modeling and extending the optimization criteria to include strain-based or damage-based measures to further enhance clinical relevance.

Conclusions

This research implemented a novel multi-femur shape optimization approach where the nail was simultaneously optimized in 10 representative femurs, selected to cover the geometric diversity of the femur population. This approach was utilized to design a long cephalomedullary nail that conforms to the curvature of the Thai medullary canal. The multi-femur shape-optimized nail can reduce stress occurring in these representative femurs.

The stresses induced by the multi-femur shape-optimized nail compared to commercial nails with radii of 1275 mm, 1500 mm, and 2000 mm in another set of 10 randomly selected femurs were validated. The multi-femur shape-optimized nail induced stress levels close to zero and reduced the average maximum principal stress of the selected femurs by 99.86%, 99.93%,

and 99.96%, respectively. These stress reductions are statistically significant (Wilcoxon signed-rank, $p = 0.036$, 0.009 , and 0.005 , respectively).

Verification under physiological loading further confirmed the mechanical robustness of the optimized design. The multi-femur shape-optimized nail exhibited substantially lower stresses around the distal screw hole (3.67 MPa) and junction (23.84 MPa) compared with commercial designs (91.68–203.26 MPa and 82.73–254.78 MPa, respectively). These results demonstrate that the optimized nail not only minimizes insertion-induced residual stress but also provides improved load transfer and reduced stress concentration in the nail under physiological conditions.

Acknowledgements

The Scholarship from the Graduate School, Chulalongkorn University, to commemorate the 72nd anniversary of His Majesty King Bhumibol Aduladej, is gratefully acknowledged.

Funding

The authors received no funding for this study.

Data availability statement

The data supporting this study's findings are available from the corresponding author (C.V.) upon reasonable request.

Additional Information

Correspondence and requests for materials should be addressed to C.V.

References

- 1 Walter, N. *et al.* Epidemiology and treatment of proximal femoral fractures in the elderly US population. *Scientific reports* 13, 12734 (2023). <https://doi.org:10.1038/s41598-023-40087-8>
- 2 Tuxun, A. *et al.* Effects of intramedullary nailing fixation on quality of life and joint function in patients with femoral neck fractures. *American Journal of Translational Research* 13, 11860 (2021).
- 3 Burnei, C., Popescu, G., Barbu, D. & Capraru, F. Intramedullary osteosynthesis versus plate osteosynthesis in subtrochanteric fractures. *Journal of medicine and life* 4, 324 (2011).
- 4 Miedel, R., Ponzer, S., Törnkvist, H., Söderqvist, A. & Tidermark, J. The standard Gamma nail or the Medoff sliding plate for unstable trochanteric and subtrochanteric fractures: a randomised, controlled trial. *The Journal of Bone & Joint Surgery British Volume* 87, 68-75 (2005).
- 5 Wang, J., Li, H., Jia, H. & Ma, X. Intramedullary versus extramedullary fixation in the treatment of subtrochanteric femur fractures: A comprehensive systematic review and meta-analysis. *Acta Orthopaedica et Traumatologica Turcica* 54, 639 (2020). <https://doi.org:10.5152/j.aott.2020.19216>
- 6 Synthes, D. *PFNA. Proximal Femoral Nail Antirotation Surgical Technique.* (2015).

- 7 Stryker. *Gamma3 Long Nail R1.5 and R2.0 Operative technique*. (2014).
- 8 Nephew, S. *TRIGEN INTERTAN Intertrochanteric Antegrade Nail*.
- 9 Chantarapanich, N., Sitthiseripratip, K., Mahaisavariya, B., Wongcumchang, M. & Siribodhi, P. 3D geometrical assessment of femoral curvature: a reverse engineering technique. *Journal of the medical association of Thailand* 91, 1377 (2011).
- 10 Cho, H.-J., Kwak, D.-S. & Kim, I.-B. Morphometric evaluation of Korean femurs by geometric computation: comparisons of the sex and the population. *BioMed Research International* 2015 (2015).
<https://doi.org/10.1155/2015/730538>
- 11 Su, X.-Y. *et al.* Three-dimensional analysis of the curvature of the femoral canal in 426 Chinese femurs. *BioMed research international* 2015 (2015). <https://doi.org/10.1155/2015/318391>
- 12 BIOMET, Z. *Zimmer Natural Nail System, Cephalomedullary Asia Nail, Surgical Technique*. 3 edn, (2016).
- 13 Shetty, A., Shenoy, P. & Swaminathan, R. Mismatch of long Gamma intramedullary nail with bow of the femur: Does radius of curvature of the nail increase risk of distal femoral complications? *Journal of clinical orthopaedics and trauma* 10, 302-304 (2019). <https://doi.org/10.1016/j.jcot.2017.12.006>
- 14 Kanawati, A. J., Jang, B., McGee, R. & Sungaran, J. The influence of entry point and radius of curvature on femoral intramedullary nail position in the distal femur. *Journal of Orthopaedics* 11, 68-71 (2014).
<https://doi.org/10.1016/j.jor.2014.04.010>
- 15 Ostrum, R. F. & Levy, M. S. Penetration of the Distal Femoral Anterior Cortex During Intramedullary Nailing for Subtrochanteric Fractures: A Report of Three Cases. *Journal of Orthopaedic Trauma* 19, 656-660 (2005).
<https://doi.org/10.1097/01.bot.0000154481.46693.69>
- 16 Egol, K. A., Chang, E. Y., Cvitkovic, J., Kummer, F. J. & Koval, K. J. Mismatch of current intramedullary nails with the anterior bow of the femur. *Journal of orthopaedic trauma* 18, 410-415 (2004). <https://doi.org/10.1097/00005131-200408000-00003>
- 17 Akman, A., Demirkan, A. F. & Akkoyunlu, N. S. Three-dimensional geometrical modelling of the femoral intramedullary cavity using ring cyclide model. *Joint diseases and related surgery* 31, 14 (2020).
<https://doi.org/10.5606/ehc.2020.65258>

- 18 Ren, D. *et al.* An innovative method for measuring the femoral arch. *Medicine* 98 (2019).
<https://doi.org/10.1097/md.00000000000015073>
- 19 Arnone, J. C. *et al.* Variability of human femoral geometry and its implications on nail design. *Injury* 52, 109-116 (2021). <https://doi.org/10.1016/j.injury.2020.09.028>
- 20 Harma, A., Germen, B., Karakas, H. M., Elmali, N. & Inan, M. The comparison of femoral curves and curves of contemporary intramedullary nails. *Surgical and radiologic anatomy* 27, 502-506 (2005).
<https://doi.org/10.1007/s00276-005-0019-2>
- 21 Abdelaal, A. H. K. *et al.* Radiological assessment of the femoral bowing in Japanese population. *SICOT-J* 2 (2016).
<https://doi.org/10.1051%2Fsicotj%2F2015037>
- 22 Tang, W., Chiu, K., Kwan, M., Ng, T. & Yau, W. Sagittal bowing of the distal femur in Chinese patients who require total knee arthroplasty. *Journal of Orthopaedic Research* 23, 41-45 (2005).
<https://doi.org/10.1016/j.orthres.2004.06.013>
- 23 Chanda, S., Gupta, S. & Kumar Pratihari, D. A genetic algorithm based multi-objective shape optimization scheme for cementless femoral implant. *Journal of Biomechanical Engineering* 137, 034502 (2015).
<https://doi.org/10.1115/1.4029061>
- 24 Korunovic, N. *et al.* In silico optimization of femoral fixator position and configuration by parametric CAD model. *Materials* 12, 2326 (2019). <https://doi.org/10.3390/ma12142326>
- 25 Carbofix. *Trochanteric (PF) Nailing System, Surgical Technique*. 9.1.2019 edn, (2019).
- 26 Chantarapanich, N. *et al.* 3D CAD/reverse engineering technique for assessment of Thai morphology: Proximal femur and acetabulum. *Journal of Orthopaedic Science* 22, 703-709 (2017). <https://doi.org/10.1016/j.jos.2017.02.003>
- 27 Jain, A. K. Data clustering: 50 years beyond K-means. *Pattern recognition letters* 31, 651-666 (2010).
<https://doi.org/10.1016/j.patrec.2009.09.011>
- 28 Saxena, A. *et al.* A review of clustering techniques and developments. *Neurocomputing* 267, 664-681 (2017).
<https://doi.org/10.1016/j.neucom.2017.06.053>
- 29 Chougule, V., Mulay, A. & Ahuja, B. Clinical case study: spine modeling for minimum invasive spine surgeries (MISS) using rapid prototyping. *Bone (CT)* 226, 3071 (2018).

- 30 Murphy, W., Black, J. & Hastings, G. W. *Handbook of biomaterial properties*. Vol. 676 (Springer, 2016).
- 31 Jang, C.-Y. *et al.* Effect of fracture levels on the strength of bone-implant constructs in subtrochanteric fracture models fixed using short cephalomedullary nails: a finite element analysis. *Injury* 50, 1883-1888 (2019).
<https://doi.org:10.1016/j.injury.2019.08.014>
- 32 Wald, A. in *Breakthroughs in Statistics: Foundations and Basic Theory* 342-357 (Springer, 1950).
- 33 Xie, W. *et al.* Comparison of three internal fixation constructs for AO/OTA 33-A3 distal femoral fractures: a biomechanical study. *Bioengineering* 11, 1110 (2024). <https://doi.org:10.3390/bioengineering11111110>
- 34 Samiezadeh, S., Avval, P. T., Fawaz, Z. & Bougherara, H. Biomechanical assessment of composite versus metallic intramedullary nailing system in femoral shaft fractures: a finite element study. *Clinical Biomechanics* 29, 803-810 (2014). <https://doi.org:10.1016/j.clinbiomech.2014.05.010>
- 35 Woods, J. C. & Della Rocca, G. J. in *Essential Biomechanics for Orthopedic Trauma: A Case-Based Guide* 221-235 (Springer, 2020).

Author contributions statement

D.S., C.V., and P.T. were responsible for the research's conceptual idea and detailed design. D.S., C.V., and P.T. designed the simulation protocol and analyzed the simulation results. D.S., C.V., and P.T. interpreted the results. D.S. wrote the first draft of the manuscript, and P.T. and C.V. provided critical revisions. All authors reviewed and approved the final manuscript.